

Project Initialization and Planning Phase

Date	29 June 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Problem Statement

Banks and financial institutions receive a large number of credit card applications every day. Evaluating these applications manually is a time-consuming process and is prone to human errors and inconsistencies. Approval decisions depend on multiple factors such as an applicant's income, employment status, credit history, outstanding debts, and financial behavior, making the assessment process complex. As a result, eligible applicants may face delays, while high-risk applicants may sometimes be approved, leading to financial losses.

The objective of this project is to develop a **Machine Learning-based Credit Card Approval Prediction System** that automates the approval process by analyzing applicant data and predicting whether a credit card application is likely to be approved or rejected. The system aims to improve decision accuracy, reduce processing time, minimize human bias, and assist banks in making faster and more reliable credit approval decisions through an easy-to-use web application.

Expected Outcome:

A secure, scalable, and accurate prediction system that enables banks to streamline credit card approval decisions while providing applicants with quick eligibility assessments.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A customer applying for a credit card.	Find out whether I am eligible for a credit card.	I don't know if I meet the bank's approval criteria.	Banks evaluate multiple financial and credit-related factors before approving	Uncertain and anxious about my application outcome.
PS-2	A bank credit analyst.	Process credit card applications quickly and accurately.	Manually reviewing every application takes a lot of time and effort.	Each application contains numerous financial and personal details that must be verified.	Overloaded and concerned about delays and errors.
PS-3	A bank risk manager.	Identify high-risk applicants before approving credit cards.	It is difficult to assess risk accurately using only manual evaluation.	Credit risk depends on several financial indicators and historical repayment	Concerned about approving risky applicants and increasing financial losses.
PS-4	A prospective customer.	Check my eligibility before submitting a formal application.	I may apply without knowing my chances of approval.	I am unaware of how my income, employment status, and credit history affect the bank's decision.	Frustrated when my application gets rejected unexpectedly.